MID-TERMEXAMINATION
(CBCS)
B. Tech (CSAI), 2nd Semester
(June, 2022)

Subject Code: BAI 102	**Subject: Object Oriented Programming Using Java**

Time : 1 ½ Hours **Maximum Marks : 30**

Note:Q. 1 is compulsory. Attempt any two questions from the rest.

Q1 **(2.5*4)**

(a) Find the 5 errors in the following code

```
package java-exam;
public class Test Programs{
        static void main (String [10] args){
                String s = "abc"
                System.out.println(s);
            }
    }
```

(b) What is multiple-inheritance? Is multiple-inheritance allowed in java? How yes/Why not?

(c) Differentiate between pure and ad-hoc polymorphism.

(d) What will be the output for following code? Give reason

```
public class Test{
        public static void main(){
                System.out.print("Hello");
                int a=10, b=20;
                System.out.print(a==b);
                System.out.print("a = " + a + b );
            }
             public static void main(String [] args){
                    System.out.print("Namastey!");
                    int a=10, b=20;
                    System.out.print("a==b");
                    System.out.print("a = " + a + b );
             }
```

Q2 **(5,5)**

(a) WAP to remove all vowels from a string entered by the user and print it. Eg. Input: "JAVA EXAM" Output: "JV XM".

(b) WAP to print reverse pyramid pattern as shown (Star & spaces)

```
* * * * *
 * * * *
  * * *
   * *
    *
```

Q3 **(10)**

(a) WAP to create an abstract class **Animals** having Sound, name as variables and read(), print() methods. Inherit the class in **Cat** (color), **Dog** (breed, height), and **Duck** (food_type). Create the 2-2 objects for each type of animals. Read functions should read details from user and print functions should print them. All classes have read() and print() without any arguments. Which type of polymorphism is this?

Q4 **(10)**

(a) Differentiate between constructor overloading and constructor chaining within same class. Explain both for the example with a Box class having length, width and height. Create constructors to 1) initialize all variables to 0, 2) all variables to user entered values, and 3) all variables to the same user entered value (cube).
Print the volume of the box for 3 different objects.